Graphs_and_tables_output

Shah Md. Tasrif Rahman

```
# Libraries ----

library(readr)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(readr)
library(tidyr)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v forcats   1.0.1     v purrr     1.2.2
v ggplot2   4.0.3     v stringr   1.6.0
v lubridate 1.9.5     v tibble    3.3.1

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```r
library(ggplot2)
library(stringr)
library(scales)
```

```
Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```r
library(writexl)
library(openxlsx)
library(e1071)
```

```
Attaching package: 'e1071'

The following object is masked from 'package:ggplot2':

    element
```

```r
library(ggpmisc)
```

```
Loading required package: ggpp
Registered S3 methods overwritten by 'ggpp':
  method                  from
  heightDetails.titleGrob ggplot2
  widthDetails.titleGrob  ggplot2

Attaching package: 'ggpp'

The following object is masked from 'package:ggplot2':

    annotate
```

```
library(gtsummary)
library(performance)
library(gt)
library(tibble)
library(tinytable)
library(flextable)
```

```
Attaching package: 'flextable'

The following object is masked from 'package:gtsummary':

    continuous_summary

The following object is masked from 'package:purrr':

    compose
```

```
library(modelsummary)
library(webshot2)
library(parameters)
```

```
Attaching package: 'parameters'

The following object is masked from 'package:modelsummary':

    supported_models

The following objects are masked from 'package:e1071':

    kurtosis, skewness
```

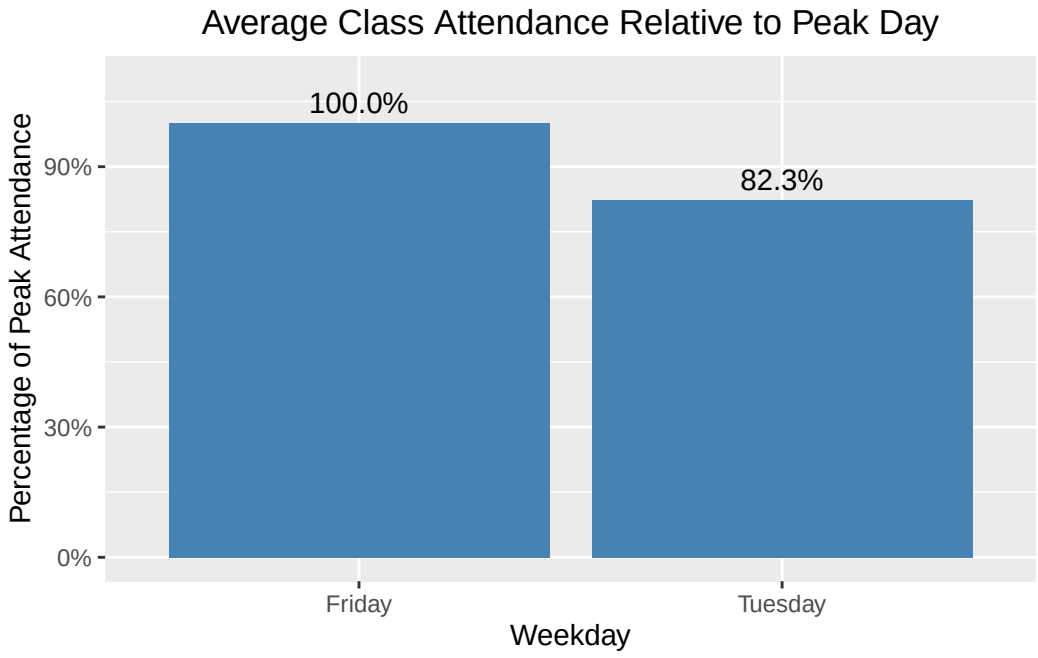
```
# Plotting ----

# Lemme just quickly load a few datasets before using them

weekday_summary <- read.xlsx(r"(E:\Statistical Softwares\R\RStudio\Projects\R_Attendance_CUSP
daily_summary <- read.xlsx(r"(E:\Statistical Softwares\R\RStudio\Projects\R_Attendance_CUSPC_
pseudo_master_dataset <- read.xlsx(r"(E:\Statistical Softwares\R\RStudio\Projects\R_Attendanc
```

```r
# 1st plot: Relative percentages between weekdays ----

ggplot(
  weekday_summary,
  aes(x = Date_Name, y = Mean_Participants / max(Mean_Participants))
) +
  geom_col(fill = "steelblue") +
  geom_text(
    aes(label = percent(Mean_Participants / max(Mean_Participants), accuracy = 0.1)),
    vjust = -0.5
  ) +
  scale_y_continuous(labels = percent, limits = c(0, 1.1)) +
  labs(
    title = "Average Class Attendance Relative to Peak Day",
    x = "Weekday",
    y = "Percentage of Peak Attendance"
  )+
  theme(plot.title = element_text(hjust = 0.5))
```

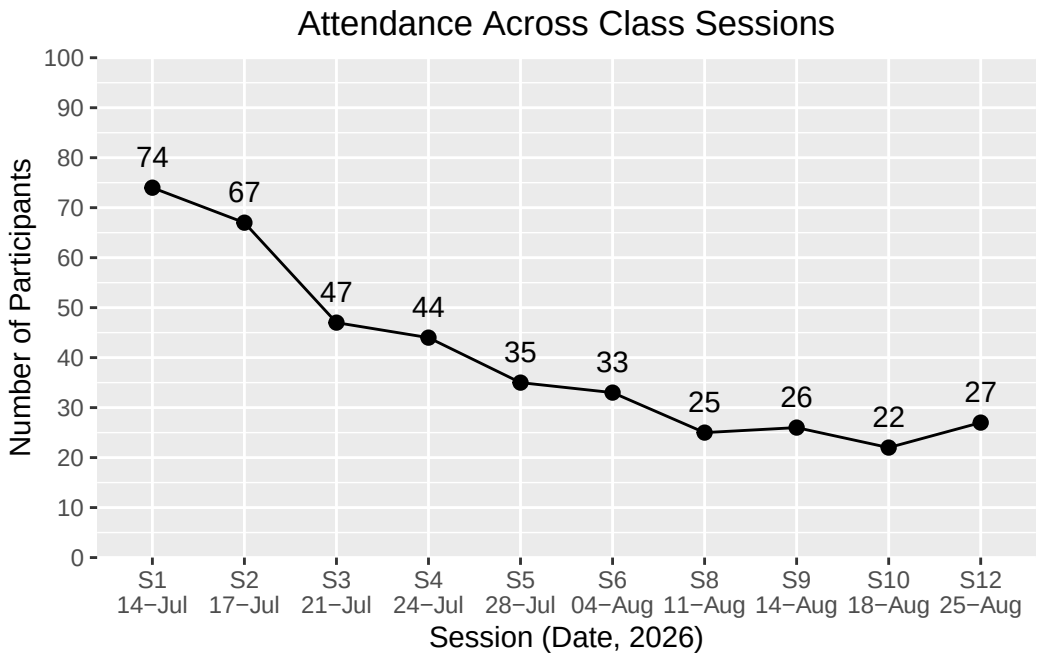



```r
# 2nd plot: Class attendance line plot ----

daily_summary$Session <- factor(
```

```
  c("S1\n14-Jul",
    "S2\n17-Jul",
    "S3\n21-Jul",
    "S4\n24-Jul",
    "S5\n28-Jul",
    "S6\n04-Aug",
    "S8\n11-Aug",
    "S9\n14-Aug",
    "S10\n18-Aug",
    "S12\n25-Aug"),
  levels = c("S1\n14-Jul",
             "S2\n17-Jul",
             "S3\n21-Jul",
             "S4\n24-Jul",
             "S5\n28-Jul",
             "S6\n04-Aug",
             "S8\n11-Aug",
             "S9\n14-Aug",
             "S10\n18-Aug",
             "S12\n25-Aug")
)

ggplot(daily_summary, aes(x = Session, y = Participants, group = 1)) +
  geom_line() +
  geom_point(size = 2.2) +
  geom_text(aes(label = Participants), vjust = -1) +
  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, by = 10),
    expand = c(0, 0)
  ) +
  labs(
    title = "Attendance Across Class Sessions",
    x = "Session (Date, 2026)",
    y = "Number of Participants"
  ) +
  theme(
    plot.title = element_text(hjust = 0.5)
  )
```




```r
# 3rd plot: Attendance histogram ----

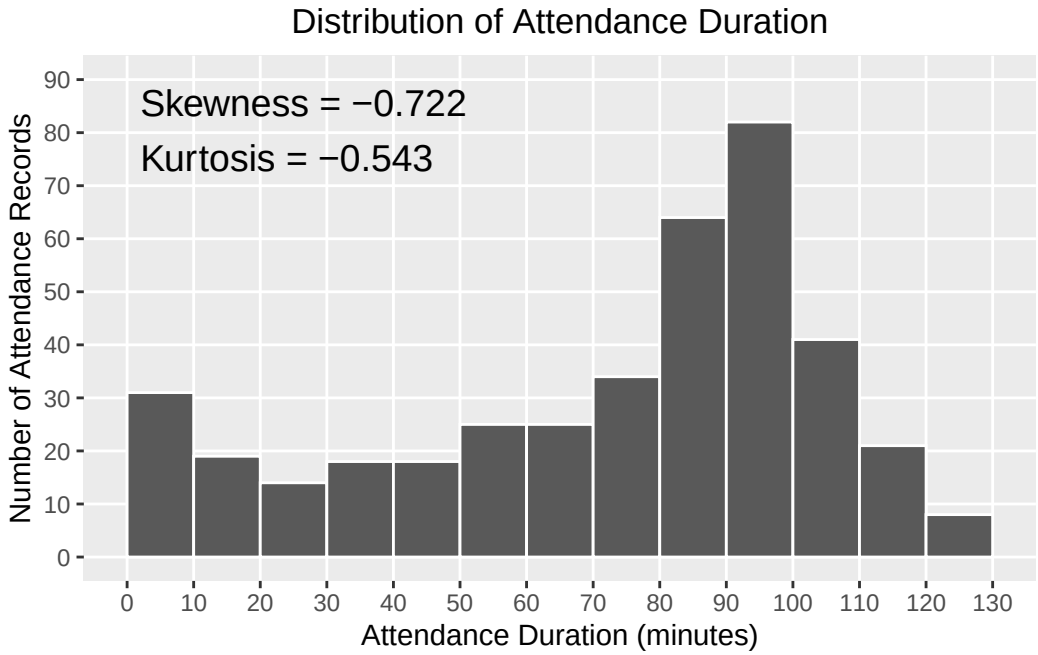
histogram_data <- pseudo_master_dataset %>%
  mutate(
    Duration_Bin = cut(
      Duration_Minutes,
      breaks = seq(0, 130, by = 10),
      right = FALSE,
      include.lowest = TRUE
    )
  ) %>%
  count(Duration_Bin, .drop = FALSE) %>%
  mutate(
    Bin_Midpoint = seq(5, 125, by = 10)
  )

skew_histogram <- e1071::skewness(pseudo_master_dataset$Duration_Minutes, type = 2)
kurt_histogram <- e1071::kurtosis(pseudo_master_dataset$Duration_Minutes, type = 2)

equation_text <- paste0(
  "Skewness = ",
  round(skew_histogram, 3),
  "\nKurtosis = ",
```

```
  round(kurt_histogram, 3)
)

ggplot(pseudo_master_dataset, aes(x = Duration_Minutes)) +
  geom_histogram(
    breaks = seq(0, 130, by = 10),
    closed = "left",
    color = "white",
    linewidth = 0.5
  ) +
  scale_x_continuous(
    breaks = seq(0, 130, by = 10),
    limits = c(0, 130),
    minor_breaks = NULL
  ) +
  scale_y_continuous(
    breaks = seq(0, ceiling(max(histogram_data$n) / 10) * 10, by = 10),
    limits = c(0, ceiling(max(histogram_data$n) / 10) * 10),
    minor_breaks = NULL
  ) +
  annotate(
    "text",
    x = 2,
    y = ceiling(max(histogram_data$n) / 10) * 10 * 0.98,
    label = equation_text,
    hjust = 0,
    vjust = 1,
    size = 5
  ) +
  labs(
    title = "Distribution of Attendance Duration",
    x = "Attendance Duration (minutes)",
    y = "Number of Attendance Records"
  ) +
  theme(plot.title = element_text(hjust = 0.5))
```




```r
# Fitting regression models ----

daily_summary$t <- as.numeric(
  daily_summary$Date - min(daily_summary$Date)
)

# Model A: Linear Regression ----

model_linear <- lm(
  Participants ~ t,
  data = daily_summary
)

# Create smooth linear fitted line

linear_curve <- data.frame(
  t = seq(
    min(daily_summary$t),
    max(daily_summary$t),
    length.out = 100
  )
)
```

```r
linear_curve$Participants <- predict(
  model_linear,
  newdata = linear_curve
)

# Model B: Exponential Decay ----

model_exponential <- nls(
  Participants ~ a * exp(b * t),
  data = daily_summary,
  start = list(
    a = max(daily_summary$Participants),
    b = -0.05
  )
)

# Model C: Asymptotic Exponential ----

model_asymptotic <- nls(
  Participants ~ c + a * exp(b * t),
  data = daily_summary,
  start = list(
    c = 25,
    # c = 25; since this is the mean of the last four somewhat similar attendance values
    a = 49,
    # a is just N(0)=74=c+a
    b = -0.05
  )
)

# Comparison table ----

model_list <- list( "Linear" = model_linear, "Exponential Decay" = model_exponential, "Asympt

# Putting pseudo R square in the model

pseudo_r2 <- function(model, y){
  rss <- sum(residuals(model)^2)
  tss <- sum((y - mean(y))^2)
  1 - rss/tss
}
```

```r
# Fit statistics
gof_custom <- tibble(
  term = c(
    "Observation Number",
    "R²",
    "Adj. R²",
    "AIC",
    "BIC",
    "RMSE"
  ),
  Linear = c(
    nobs(model_linear),
    summary(model_linear)$r.squared,
    summary(model_linear)$adj.r.squared,
    AIC(model_linear),
    BIC(model_linear),
    sqrt(mean(residuals(model_linear)^2))
  ),
  `Exponential Decay` = c(
    nobs(model_exponential),
    pseudo_r2(model_exponential, daily_summary$Participants),
    NA,
    AIC(model_exponential),
    BIC(model_exponential),
    sqrt(mean(residuals(model_exponential)^2))
  ),
  `Asymptotic Exponential` = c(
    nobs(model_asymptotic),
    pseudo_r2(model_asymptotic, daily_summary$Participants),
    NA,
    AIC(model_asymptotic),
    BIC(model_asymptotic),
    sqrt(mean(residuals(model_asymptotic)^2))
  )
)

comparison_table <- modelsummary(
  model_list,
  output = "flextable",
  statistic = NULL,
  gof_omit = ".*",
  add_rows = gof_custom
```

```
)

comparison_table <- colformat_double(
  comparison_table,
  digits = 3
)

comparison_table <- bg(
  comparison_table,
  bg = "white",
  part = "all"
)

save_as_image(
  comparison_table,
  path = r"(E:\Statistical Softwares\R\RStudio\Projects\R_Attendance_CUSPC_SPSS_2026\Graphs_a
  expand = 0,
  zoom = 2,
  overwrite = TRUE
)
```

```
[1] "E:\\Statistical Softwares\\R\\RStudio\\Projects\\R_Attendance_CUSPC_SPSS_2026\\Graphs_ar
```

```
# Parameters among models ----

model_parameters(model_linear)
```

```
Parameter   | Coefficient |   SE |         95% CI |  t(8) |      p
-----------------------------------------------------------------
(Intercept) |       61.09 | 4.87 | [49.87, 72.31] | 12.55 | < .001
t           |       -1.10 | 0.21 | [-1.58, -0.63] | -5.34 | < .001


Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
  using a Wald t-distribution approximation.
```

```
model_parameters(model_exponential)
```

```
# Fixed Effects
```

```
Parameter | Coefficient |       SE |         95% CI |  t(8) |      p
-------------------------------------------------------------------
a         |       68.28 |     4.17 | [58.67, 77.89] | 16.39 | < .001
b         |       -0.03 | 4.54e-03 | [-0.04, -0.02] | -7.47 | < .001

Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
  using a Wald t-distribution approximation.
```

```
model_parameters(model_asymptotic)
```

```
# Fixed Effects

Parameter | Coefficient |   SE |         95% CI |  t(7) |      p
-----------------------------------------------------------------
c         |       23.20 | 2.19 | [18.03, 28.37] | 10.61 | < .001
a         |       52.69 | 2.98 | [45.64, 59.74] | 17.67 | < .001
b         |       -0.10 | 0.01 | [-0.13, -0.06] | -6.57 | < .001

Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
  using a Wald t-distribution approximation.
```

```
# 4th plot: Linear fit ----

ggplot(
  daily_summary,
  aes(x = t, y = Participants)
) +

  # Actual attendance
  geom_line() +
  geom_point(size = 2.2) +

  # Linear fitted line
  geom_line(
    data = linear_curve,
    aes(x = t, y = Participants),
    linetype = "dotted",
    linewidth = 1,
    color = "steelblue3"
```

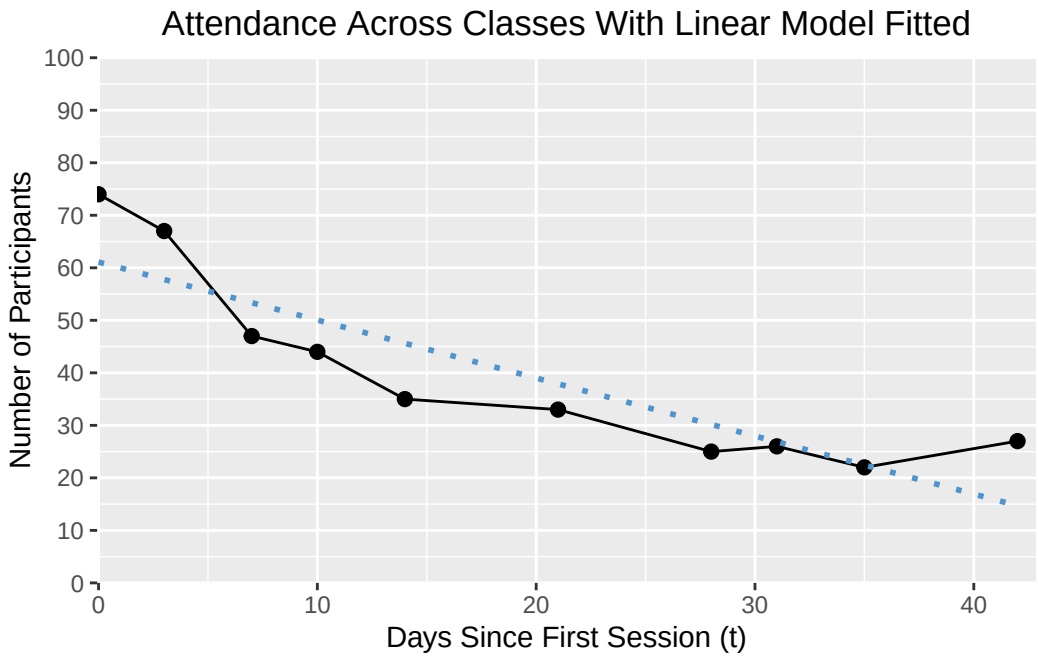
```
) +

scale_x_continuous(
  limits = c(0, max(daily_summary$t)),
  breaks = pretty(daily_summary$t),
  expand = expansion(mult = c(0.0015, 0.02))
) +

scale_y_continuous(
  limits = c(0, 100),
  breaks = seq(0, 100, by = 10),
  expand = c(0, 0)
) +

labs(
  title = "Attendance Across Classes With Linear Model Fitted",
  x = "Days Since First Session (t)",
  y = "Number of Participants"
) +

theme(
  plot.title = element_text(hjust = 0.5)
)
```




```
# 5th plot: Exponential Decay fit ----

# Create smooth exponential fitted line
exp_curve <- data.frame(
  t = seq(
    min(daily_summary$t),
    max(daily_summary$t),
    length.out = 100
  )
)

# Predicted participants
exp_curve$Participants <- predict(
  model_exponential,
  newdata = exp_curve
)

ggplot(
  daily_summary,
  aes(x = t, y = Participants)
) +

  # Actual attendance
```

```
  geom_line() +
  geom_point(size = 2.2) +

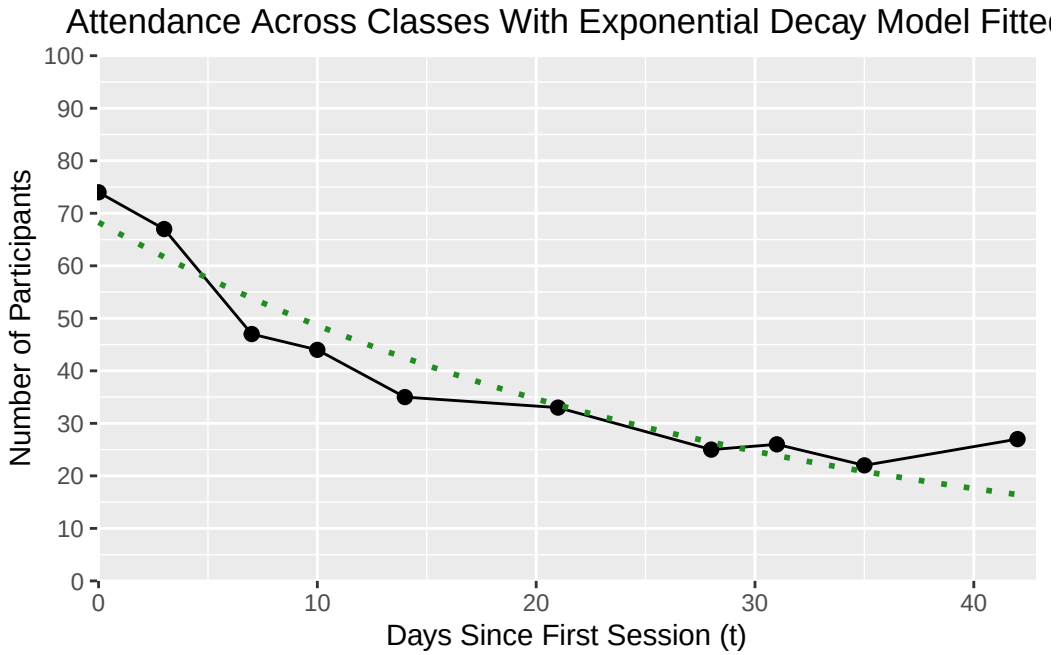
  # Exponential fitted line
  geom_line(
    data = exp_curve,
    aes(x = t, y = Participants),
    linetype = "dotted",
    linewidth = 1,
    color = "forestgreen"
  ) +

  scale_x_continuous(
    limits = c(0, max(daily_summary$t)),
    breaks = pretty(daily_summary$t),
    expand = expansion(mult = c(0.0015, 0.02))
  ) +

  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, by = 10),
    expand = c(0, 0)
  ) +

  labs(
    title = "Attendance Across Classes With Exponential Decay Model Fitted",
    x = "Days Since First Session (t)",
    y = "Number of Participants"
  ) +

  theme(
    plot.title = element_text(hjust = 0.5)
  )
```




```
# 6th plot: Asymptotic Exponential fit ----

# Create smooth asymptotic fitted line
asym_curve <- data.frame(
  t = seq(
    min(daily_summary$t),
    max(daily_summary$t),
    length.out = 100
  )
)

# Predicted participants
asym_curve$Participants <- predict(
  model_asymptotic,
  newdata = asym_curve
)

ggplot(
  daily_summary,
  aes(x = t, y = Participants)
) +

  # Actual attendance
```

```
  geom_line() +
  geom_point(size = 2.2) +

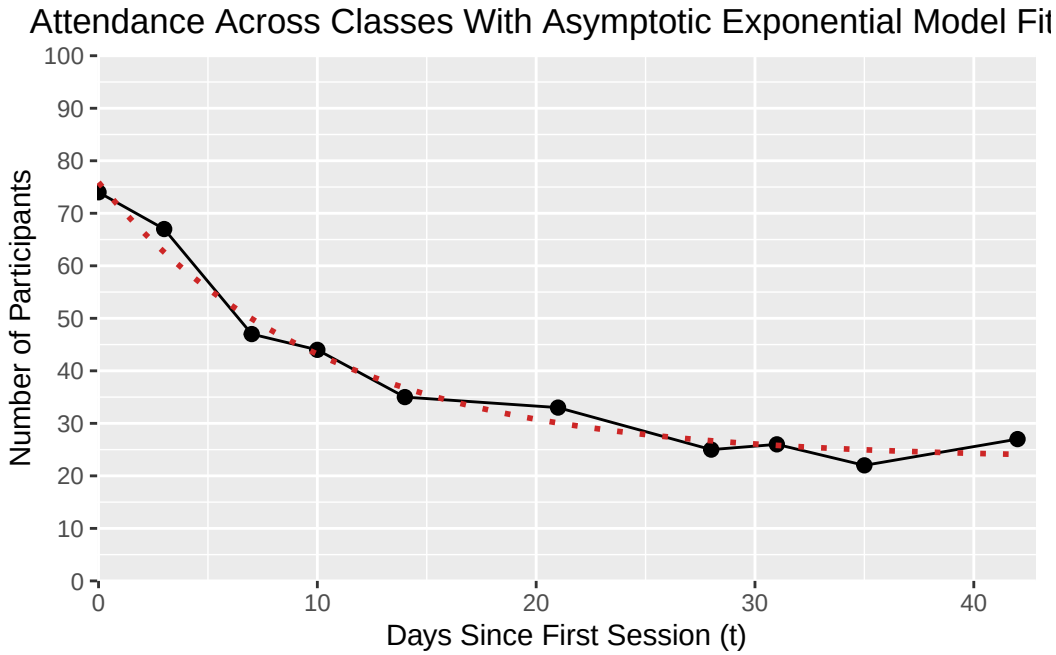
  # Asymptotic exponential fitted line
  geom_line(
    data = asym_curve,
    aes(x = t, y = Participants),
    linetype = "dotted",
    linewidth = 1,
    color = "firebrick3"
  ) +

  scale_x_continuous(
    limits = c(0, max(daily_summary$t)),
    breaks = pretty(daily_summary$t),
    expand = expansion(mult = c(0.0015, 0.02))
  ) +

  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, by = 10),
    expand = c(0, 0)
  ) +

  labs(
    title = "Attendance Across Classes With Asymptotic Exponential Model Fitted",
    x = "Days Since First Session (t)",
    y = "Number of Participants"
  ) +

  theme(
    plot.title = element_text(hjust = 0.5)
  )
```




```
# 7th plot: Model comparison ----

ggplot(
  daily_summary,
  aes(x = t, y = Participants)
) +

  # Actual attendance
  geom_line(
    color = "grey27",
    linewidth = .8
  ) +

  geom_point(
    color = "black",
    size = 2.2
  ) +

  # Linear fit
  geom_line(
    data = linear_curve,
    aes(x = t, y = Participants,
        color = "Linear"),
```

```
  linewidth = 1.1
) +

# Exponential fit
geom_line(
  data = exp_curve,
  aes(x = t, y = Participants,
      color = "Exponential"),
  linewidth = 1.1
) +

# Asymptotic exponential fit
geom_line(
  data = asym_curve,
  aes(x = t, y = Participants,
      color = "Asymptotic Exponential"),
  linewidth = 1.1
) +

scale_color_manual(
  name = "Model",
  values = c(
    "Linear" = "steelblue3",
    "Exponential" = "forestgreen",
    "Asymptotic Exponential" = "firebrick3"
  )
) +

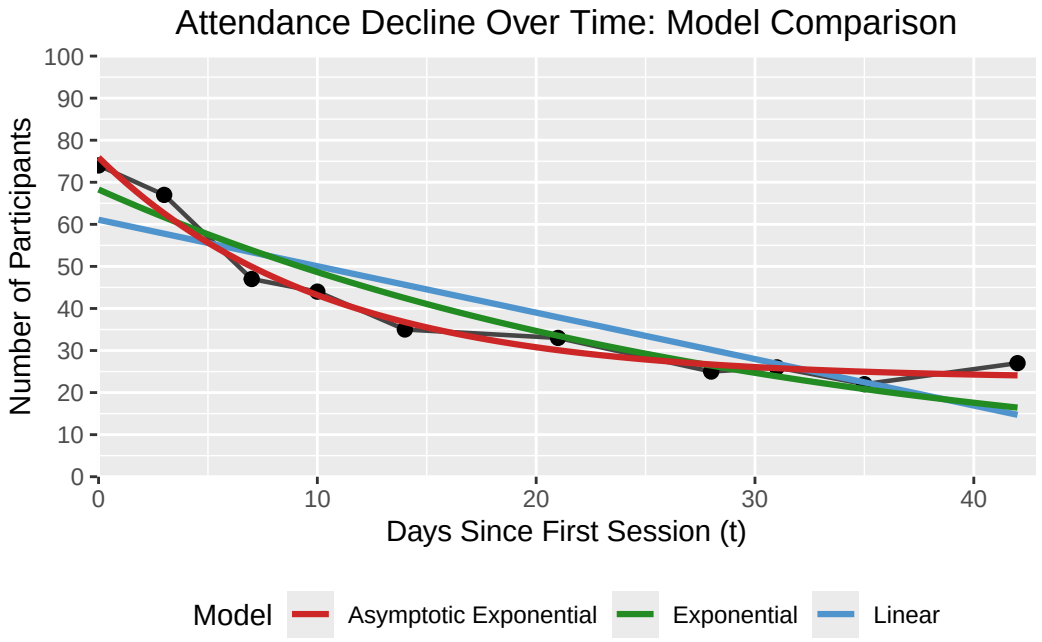
scale_x_continuous(
  limits = c(0, max(daily_summary$t)),
  breaks = pretty(daily_summary$t),
  expand = expansion(mult = c(0.0015, 0.02))
) +

scale_y_continuous(
  limits = c(0, 100),
  breaks = seq(0, 100, by = 10),
  expand = c(0, 0)
) +

labs(
  title = "Attendance Decline Over Time: Model Comparison",
```

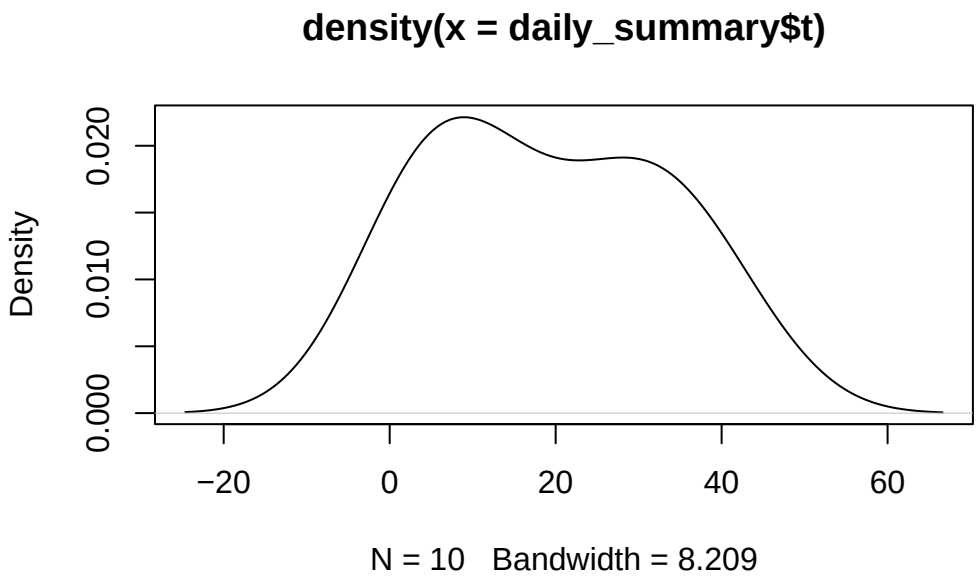
```
    x = "Days Since First Session (t)",
    y = "Number of Participants"
  ) +

  theme(
    plot.title = element_text(hjust = 0.5),
    legend.position = "bottom"
  )
```

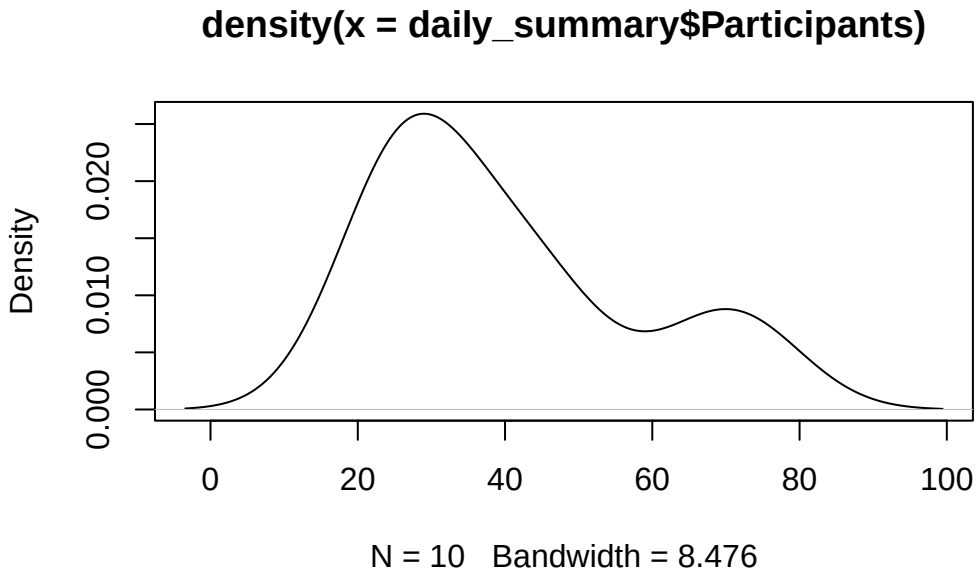



```
# Other plots and tests ----

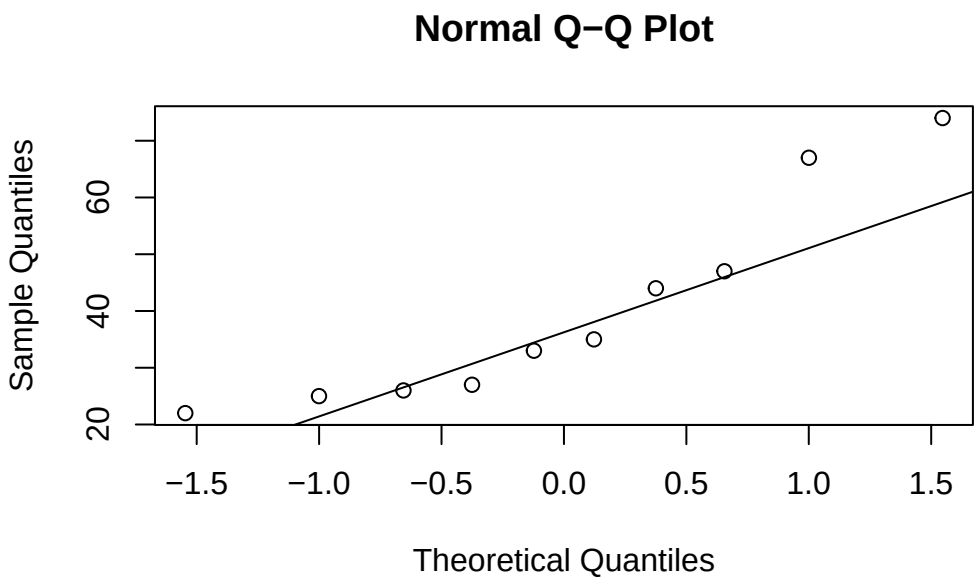
plot(density(daily_summary$t))
```




```
plot(density(daily_summary$Participants))
```



```
qqnorm(daily_summary$Participants)
qqline(daily_summary$Participants)
```




```
shapiro.test(daily_summary$Participants)
```

```
    Shapiro-Wilk normality test

data:  daily_summary$Participants
W = 0.86181, p-value = 0.08014
```

```
# Model comparison 2: Electric Boogaloo

library(broom)
glance(model_linear)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
      <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
1     0.781         0.753  8.98      28.5 0.000698     1  -35.0  76.0  76.9
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
glance(model_exponential)
```

```
# A tibble: 1 x 9
  sigma isConv     finTol logLik   AIC   BIC deviance df.residual  nobs
  <dbl> <lgl>       <dbl>  <dbl> <dbl> <dbl>    <dbl>       <int> <int>
1  6.18 TRUE   0.00000382  -31.3  68.6  69.5     305.           8    10
```

```r
glance(model_asymptotic)
```

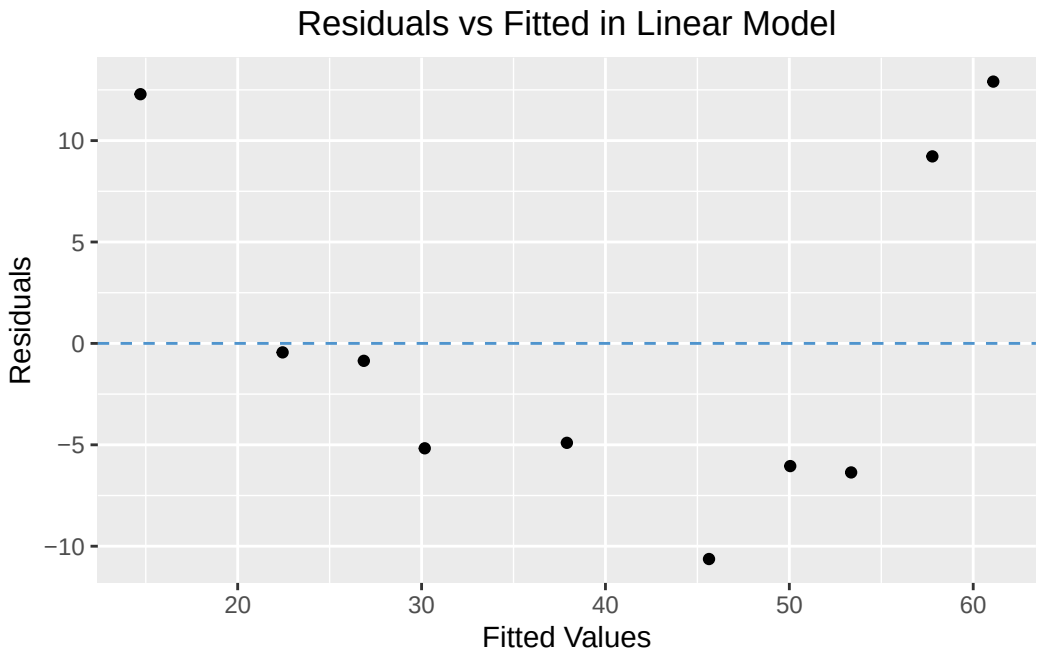
```
# A tibble: 1 x 9
  sigma isConv     finTol logLik   AIC   BIC deviance df.residual  nobs
  <dbl> <lgl>       <dbl>  <dbl> <dbl> <dbl>    <dbl>       <int> <int>
1  3.02 TRUE   0.00000825  -23.5  54.9  56.2     64.0           7    10
```

```r
# Plotting residuals vs fitted values

# Linear

res_vs_fit_linear <- data.frame(
  Fitted = fitted(model_linear),
  Residuals = residuals(model_linear)
)

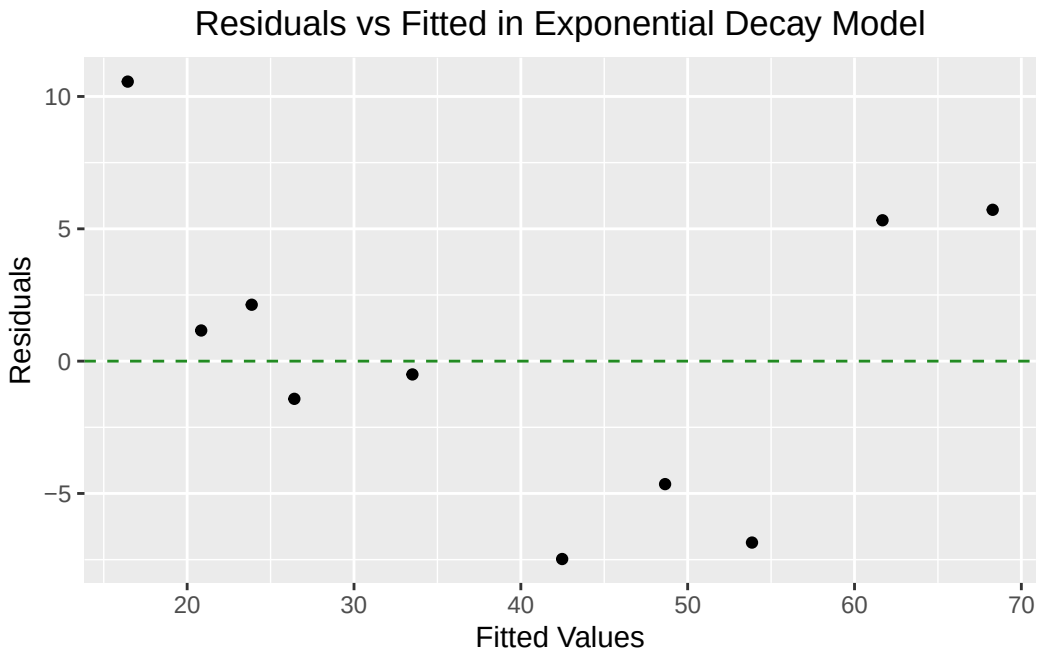
ggplot(res_vs_fit_linear, aes(x = Fitted, y = Residuals)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "steelblue3", linetype = "dashed") +
  labs(title = "Residuals vs Fitted in Linear Model", x = "Fitted Values", y = "Residuals") +
  theme(plot.title = element_text(hjust = 0.5))
```




```r
# Exponential decay

res_vs_fit_exponential <- data.frame(
  Fitted = fitted(model_exponential),
  Residuals = residuals(model_exponential)
)

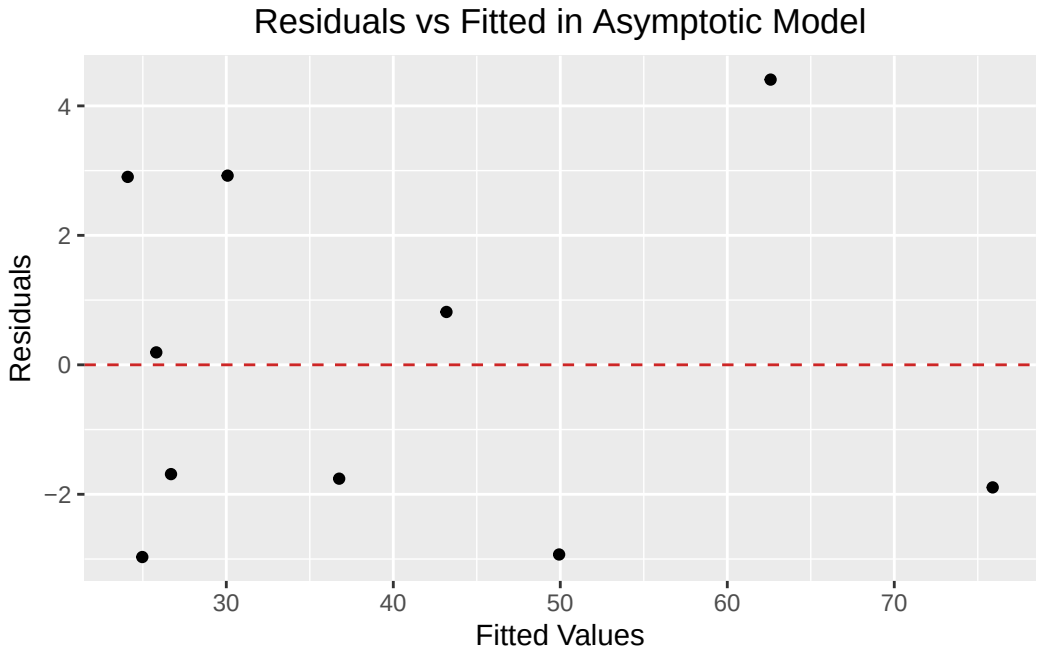
ggplot(res_vs_fit_exponential, aes(x = Fitted, y = Residuals)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "forestgreen", linetype = "dashed") +
  labs(title = "Residuals vs Fitted in Exponential Decay Model", x = "Fitted Values", y = "Re
  theme(plot.title = element_text(hjust = 0.5))
```




```r
# Asymptotic

res_vs_fit_asymptotic <- data.frame(
  Fitted = fitted(model_asymptotic),
  Residuals = residuals(model_asymptotic)
)

ggplot(res_vs_fit_asymptotic, aes(x = Fitted, y = Residuals)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "firebrick3", linetype = "dashed") +
  labs(title = "Residuals vs Fitted in Asymptotic Model", x = "Fitted Values", y = "Residuals
  theme(plot.title = element_text(hjust = 0.5))
```




```
# ANOVA between nested models

anova(model_exponential, model_asymptotic)
```

```
Analysis of Variance Table

Model 1: Participants ~ a * exp(b * t)
Model 2: Participants ~ c + a * exp(b * t)
  Res.Df Res.Sum Sq Df Sum Sq F value   Pr(>F)
1      8    305.237
2      7     64.013  1 241.22  26.379 0.001345 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Adding the session variable as explanatory/independent variable instead of t to make simila

# Model D: Linear Regression ----

daily_summary_session_number_fixed <- read.xlsx(r"(E:\Statistical Softwares\R\RStudio\Project

model_linear_snf <- lm(
  Participants ~ Session_Number,
```

```
  data = daily_summary_session_number_fixed
)

# Create smooth linear fitted line

linear_curve_snf <- data.frame(
  Session_Number = seq(
    min(daily_summary_session_number_fixed$Session_Number),
    max(daily_summary_session_number_fixed$Session_Number),
    length.out = 100
  )
)

linear_curve_snf$Participants <- predict(
  model_linear_snf,
  newdata = linear_curve_snf
)

# Model E: Exponential Decay ----

model_exponential_snf <- nls(
  Participants ~ a * exp(b * Session_Number),
  data = daily_summary_session_number_fixed,
  start = list(
    a = max(daily_summary_session_number_fixed$Participants),
    b = -0.05
  )
)

# Model F: Asymptotic Exponential ----

model_asymptotic_snf <- nls(
  Participants ~ c + a * exp(b * Session_Number),
  data = daily_summary_session_number_fixed,
  start = list(
    c = 25,
    # c = 25; since this is the mean of the last four somewhat similar attendance values
    a = 49,
    # a is just N(1)=74=c+a
    b = -0.05
  )
)
```

```r
# Comparison table ----

model_list_snf <- list( "Linear" = model_linear_snf, "Exponential Decay" = model_exponential_

# Putting pseudo R square in the model

pseudo_r2 <- function(model, y){
  rss <- sum(residuals(model)^2)
  tss <- sum((y - mean(y))^2)
  1 - rss/tss
}

# Fit statistics
gof_custom <- tibble(
  term = c(
    "Observation Number",
    "R²",
    "Adj. R²",
    "AIC",
    "BIC",
    "RMSE"
  ),
  Linear = c(
    nobs(model_linear_snf),
    summary(model_linear_snf)$r.squared,
    summary(model_linear_snf)$adj.r.squared,
    AIC(model_linear_snf),
    BIC(model_linear_snf),
    sqrt(mean(residuals(model_linear_snf)^2))
  ),
  `Exponential Decay` = c(
    nobs(model_exponential_snf),
    pseudo_r2(model_exponential_snf, daily_summary_session_number_fixed$Participants),
    NA,
    AIC(model_exponential_snf),
    BIC(model_exponential_snf),
    sqrt(mean(residuals(model_exponential_snf)^2))
  ),
  `Asymptotic Exponential` = c(
    nobs(model_asymptotic_snf),
    pseudo_r2(model_asymptotic_snf, daily_summary_session_number_fixed$Participants),
    NA,
```

```
    AIC(model_asymptotic_snf),
    BIC(model_asymptotic_snf),
    sqrt(mean(residuals(model_asymptotic_snf)^2))
  )
)

comparison_table_snf <- modelsummary(
  model_list_snf,
  output = "flextable",
  statistic = NULL,
  gof_omit = ".*",
  add_rows = gof_custom
)

comparison_table_snf <- colformat_double(
  comparison_table_snf,
  digits = 3
)

comparison_table_snf <- bg(
  comparison_table_snf,
  bg = "white",
  part = "all"
)

save_as_image(
  comparison_table_snf,
  path = r"(E:\Statistical Softwares\R\RStudio\Projects\R_Attendance_CUSPC_SPSS_2026\Graphs_a
  expand = 0,
  zoom = 2,
  overwrite = TRUE
)
```

```
[1] "E:\\Statistical Softwares\\R\\RStudio\\Projects\\R_Attendance_CUSPC_SPSS_2026\\Graphs_an
```

```
# Parameters among models ----

model_parameters(model_linear_snf)
```

```
Parameter      | Coefficient |   SE |         95% CI |  t(8) |      p
-------------------------------------------------------------------
```

```
(Intercept)    |       66.20 | 5.70 | [53.06, 79.34] | 11.61 | < .001
Session Number |       -4.37 | 0.82 | [-6.26, -2.47] | -5.31 | < .001


Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
  using a Wald t-distribution approximation.
```

```
model_parameters(model_exponential_snf)
```

```
# Fixed Effects

Parameter | Coefficient |   SE |         95% CI |  t(8) |      p
-------------------------------------------------------------
a         |       80.32 | 5.79 | [66.97, 93.68] | 13.87 | < .001
b         |       -0.14 | 0.02 | [-0.17, -0.10] | -8.01 | < .001


Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
  using a Wald t-distribution approximation.
```

```
model_parameters(model_asymptotic_snf)
```

```
# Fixed Effects

Parameter | Coefficient |   SE |         95% CI |  t(7) |      p
-------------------------------------------------------------
c         |       21.60 | 2.96 | [14.61, 28.60] |  7.30 | < .001
a         |       75.98 | 5.87 | [62.11, 89.86] | 12.95 | < .001
b         |       -0.32 | 0.06 | [-0.46, -0.19] | -5.67 | < .001


Uncertainty intervals (equal-tailed) and p-values (two-tailed) computed
  using a Wald t-distribution approximation.
```

```
# 8th plot: Linear fit ----

ggplot(
  daily_summary_session_number_fixed,
  aes(x = Session_Number, y = Participants)
```

```
) +

  # Actual attendance
  geom_line() +
  geom_point(size = 2.2) +

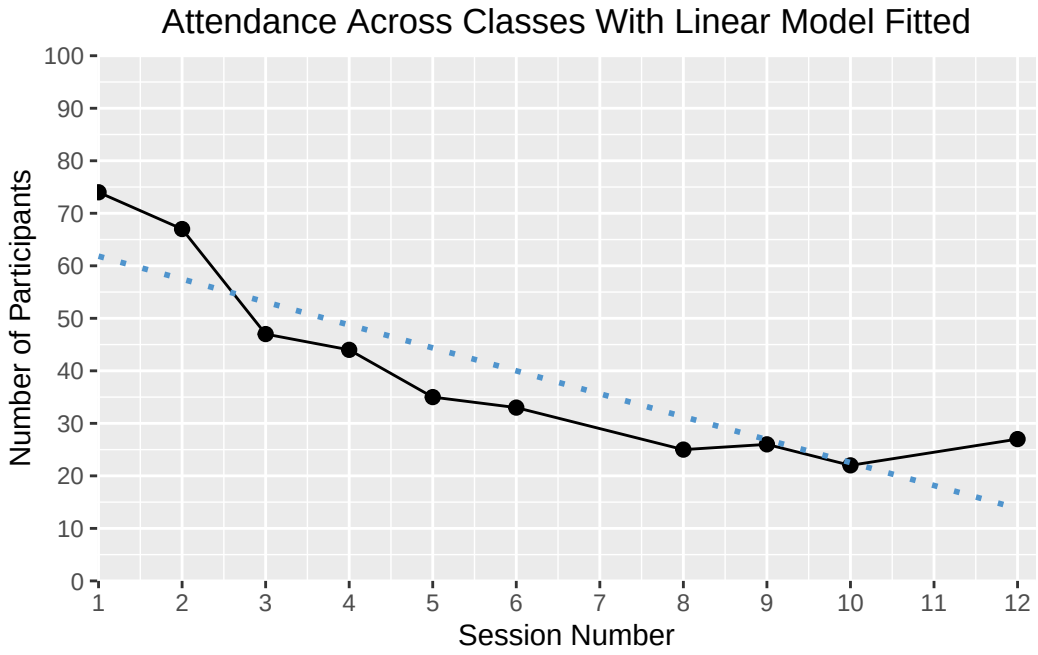
  # Linear fitted line
  geom_line(
    data = linear_curve_snf,
    aes(x = Session_Number, y = Participants),
    linetype = "dotted",
    linewidth = 1,
    color = "steelblue3"
  ) +

  scale_x_continuous(
    limits = c(1, max(daily_summary_session_number_fixed$Session_Number)),
    breaks = seq(1, max(daily_summary_session_number_fixed$Session_Number), by = 1),
    expand = expansion(mult = c(0.0015, 0.02))
  ) +

  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, by = 10),
    expand = c(0, 0)
  ) +

  labs(
    title = "Attendance Across Classes With Linear Model Fitted",
    x = "Session Number",
    y = "Number of Participants"
  ) +

  theme(
    plot.title = element_text(hjust = 0.5)
  )
```




```
# 9th plot: Exponential Decay fit ----

# Create smooth exponential fitted line
exp_curve_snf <- data.frame(
  Session_Number = seq(
    min(daily_summary_session_number_fixed$Session_Number),
    max(daily_summary_session_number_fixed$Session_Number),
    length.out = 100
  )
)

# Predicted participants
exp_curve_snf$Participants <- predict(
  model_exponential_snf,
  newdata = exp_curve_snf
)

ggplot(
  daily_summary_session_number_fixed,
  aes(x = Session_Number, y = Participants)
) +

  # Actual attendance
```

```
  geom_line() +
  geom_point(size = 2.2) +

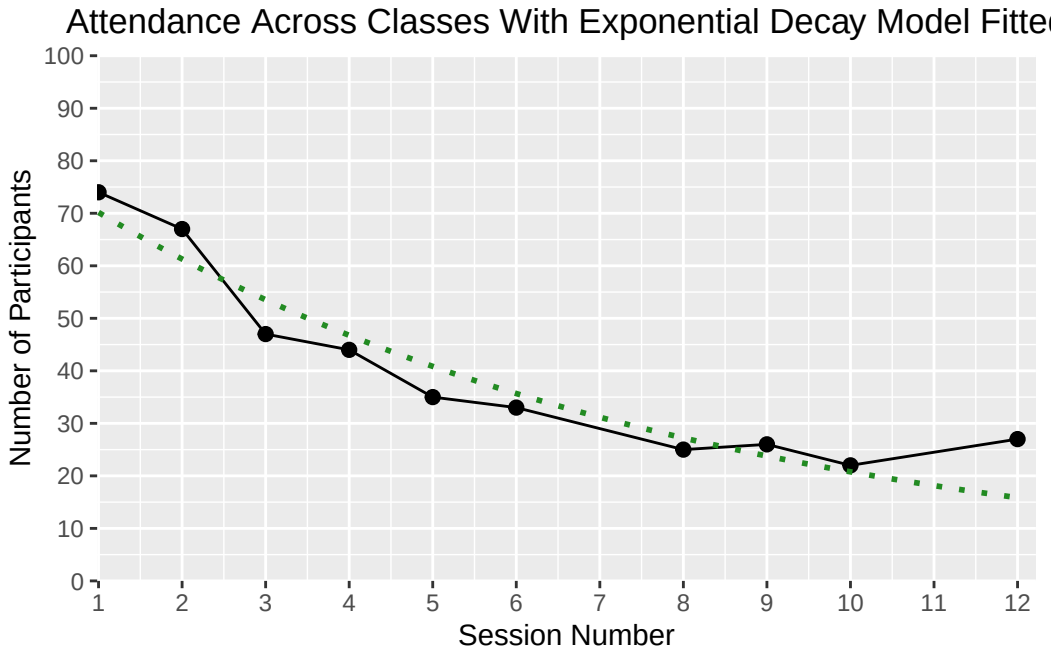
  # Exponential fitted line
  geom_line(
    data = exp_curve_snf,
    aes(x = Session_Number, y = Participants),
    linetype = "dotted",
    linewidth = 1,
    color = "forestgreen"
  ) +

  scale_x_continuous(
    limits = c(1, max(daily_summary_session_number_fixed$Session_Number)),
    breaks = seq(1, max(daily_summary_session_number_fixed$Session_Number), by = 1),
    expand = expansion(mult = c(0.0015, 0.02))
  ) +

  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, by = 10),
    expand = c(0, 0)
  ) +

  labs(
    title = "Attendance Across Classes With Exponential Decay Model Fitted",
    x = "Session Number",
    y = "Number of Participants"
  ) +

  theme(
    plot.title = element_text(hjust = 0.5)
  )
```




```
# 10th plot: Asymptotic Exponential fit ----

# Create smooth asymptotic fitted line
asym_curve_snf <- data.frame(
  Session_Number = seq(
    min(daily_summary_session_number_fixed$Session_Number),
    max(daily_summary_session_number_fixed$Session_Number),
    length.out = 100
  )
)

# Predicted participants
asym_curve_snf$Participants <- predict(
  model_asymptotic_snf,
  newdata = asym_curve_snf
)

ggplot(
  daily_summary_session_number_fixed,
  aes(x = Session_Number, y = Participants)
) +

  # Actual attendance
```

```
  geom_line() +
  geom_point(size = 2.2) +

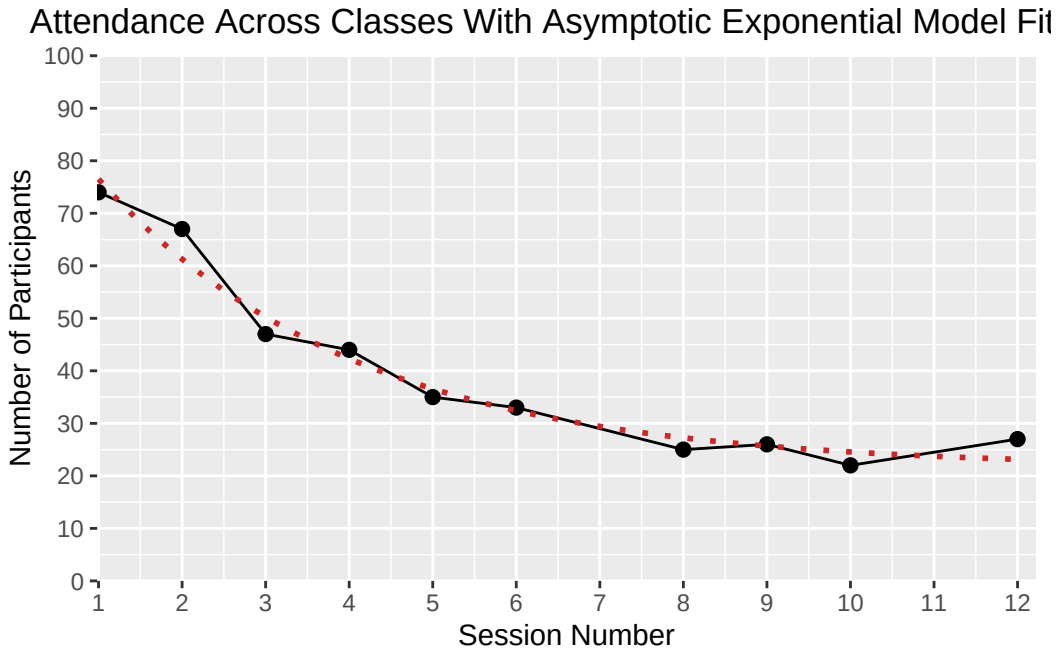
  # Asymptotic exponential fitted line
  geom_line(
    data = asym_curve_snf,
    aes(x = Session_Number, y = Participants),
    linetype = "dotted",
    linewidth = 1,
    color = "firebrick3"
  ) +

  scale_x_continuous(
    limits = c(1, max(daily_summary_session_number_fixed$Session_Number)),
    breaks = seq(1, max(daily_summary_session_number_fixed$Session_Number), by = 1),
    expand = expansion(mult = c(0.0015, 0.02))
  ) +

  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, by = 10),
    expand = c(0, 0)
  ) +

  labs(
    title = "Attendance Across Classes With Asymptotic Exponential Model Fitted",
    x = "Session Number",
    y = "Number of Participants"
  ) +

  theme(
    plot.title = element_text(hjust = 0.5)
  )
```




```
# 11th plot: Model comparison ----

ggplot(
  daily_summary_session_number_fixed,
  aes(x = Session_Number, y = Participants)
) +

  # Actual attendance
  geom_line(
    color = "grey27",
    linewidth = .8
  ) +

  geom_point(
    color = "black",
    size = 2.2
  ) +

  # Linear fit
  geom_line(
    data = linear_curve_snf,
    aes(x = Session_Number, y = Participants,
        color = "Linear"),
```

```
  linewidth = 1.1
) +

# Exponential fit
geom_line(
  data = exp_curve_snf,
  aes(x = Session_Number, y = Participants,
      color = "Exponential"),
  linewidth = 1.1
) +

# Asymptotic exponential fit
geom_line(
  data = asym_curve_snf,
  aes(x = Session_Number, y = Participants,
      color = "Asymptotic Exponential"),
  linewidth = 1.1
) +

scale_color_manual(
  name = "Model",
  values = c(
    "Linear" = "steelblue3",
    "Exponential" = "forestgreen",
    "Asymptotic Exponential" = "firebrick3"
  )
) +

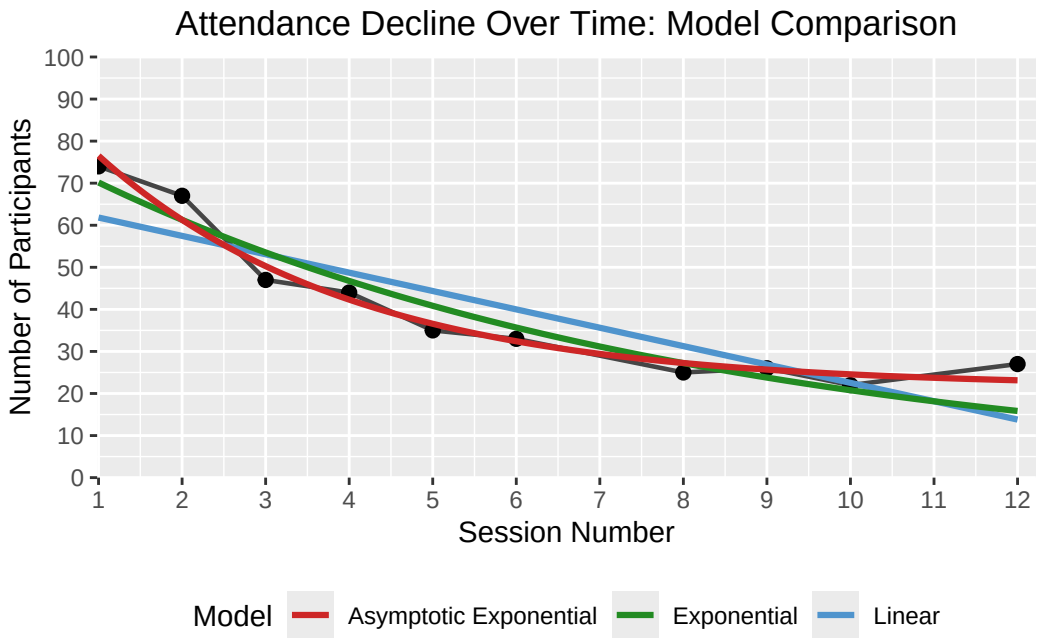
scale_x_continuous(
  limits = c(1, max(daily_summary_session_number_fixed$Session_Number)),
  breaks = seq(1, max(daily_summary_session_number_fixed$Session_Number), by = 1),
  expand = expansion(mult = c(0.0015, 0.02))
) +

scale_y_continuous(
  limits = c(0, 100),
  breaks = seq(0, 100, by = 10),
  expand = c(0, 0)
) +

labs(
  title = "Attendance Decline Over Time: Model Comparison",
```

```
    x = "Session Number",
    y = "Number of Participants"
  ) +

  theme(
    plot.title = element_text(hjust = 0.5),
    legend.position = "bottom"
  )
```




```
# Other plots and tests ----

# Model comparison 2: Electric Boogaloo pt. 2

library(broom)
glance(model_linear_snf)
```

```
# A tibble: 1 x 12
  r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC
      <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>
1     0.779         0.751  9.01      28.2 0.000722     1  -35.1  76.1  77.0
# i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
glance(model_exponential_snf)
```

```
# A tibble: 1 x 9
  sigma isConv     finTol logLik   AIC   BIC deviance df.residual  nobs
  <dbl> <lgl>       <dbl>  <dbl> <dbl> <dbl>    <dbl>       <int> <int>
1  5.86 TRUE   0.00000469  -30.8  67.5  68.4     275.           8    10
```

```r
glance(model_asymptotic_snf)
```

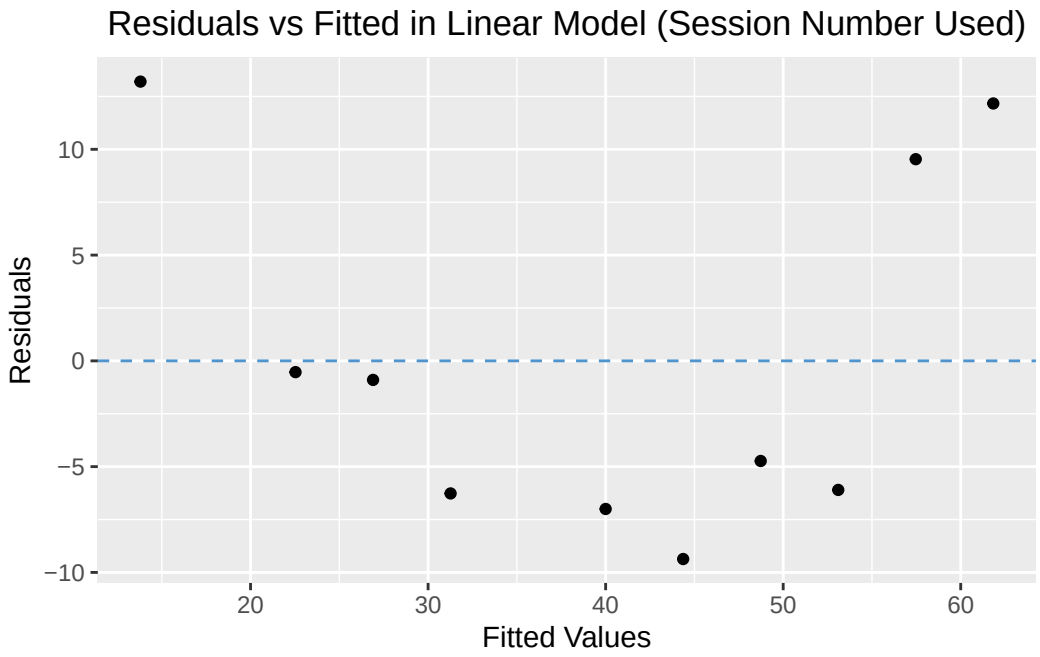
```
# A tibble: 1 x 9
  sigma isConv     finTol logLik   AIC   BIC deviance df.residual  nobs
  <dbl> <lgl>       <dbl>  <dbl> <dbl> <dbl>    <dbl>       <int> <int>
1  3.42 TRUE   0.00000447  -24.7  57.4  58.6     81.9           7    10
```

```r
# Plotting residuals vs fitted values

# Linear

res_vs_fit_linear_snf <- data.frame(
  Fitted = fitted(model_linear_snf),
  Residuals = residuals(model_linear_snf)
)

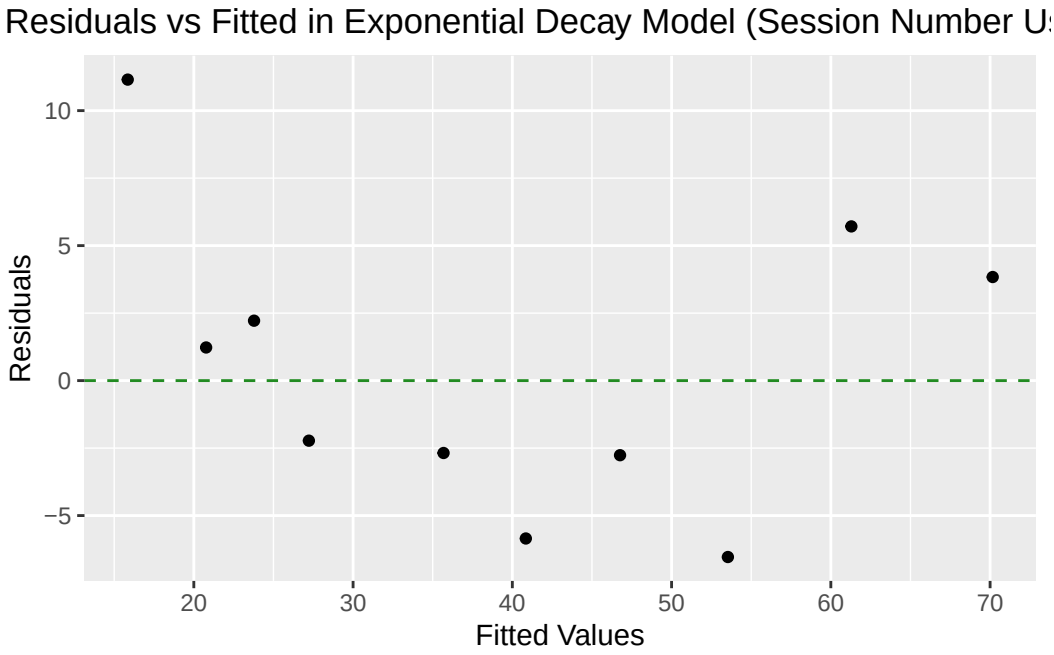
ggplot(res_vs_fit_linear_snf, aes(x = Fitted, y = Residuals)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "steelblue3", linetype = "dashed") +
  labs(title = "Residuals vs Fitted in Linear Model (Session Number Used)", x = "Fitted Valu
  theme(plot.title = element_text(hjust = 0.5))
```




```r
# Exponential decay

res_vs_fit_exponential_snf <- data.frame(
  Fitted = fitted(model_exponential_snf),
  Residuals = residuals(model_exponential_snf)
)

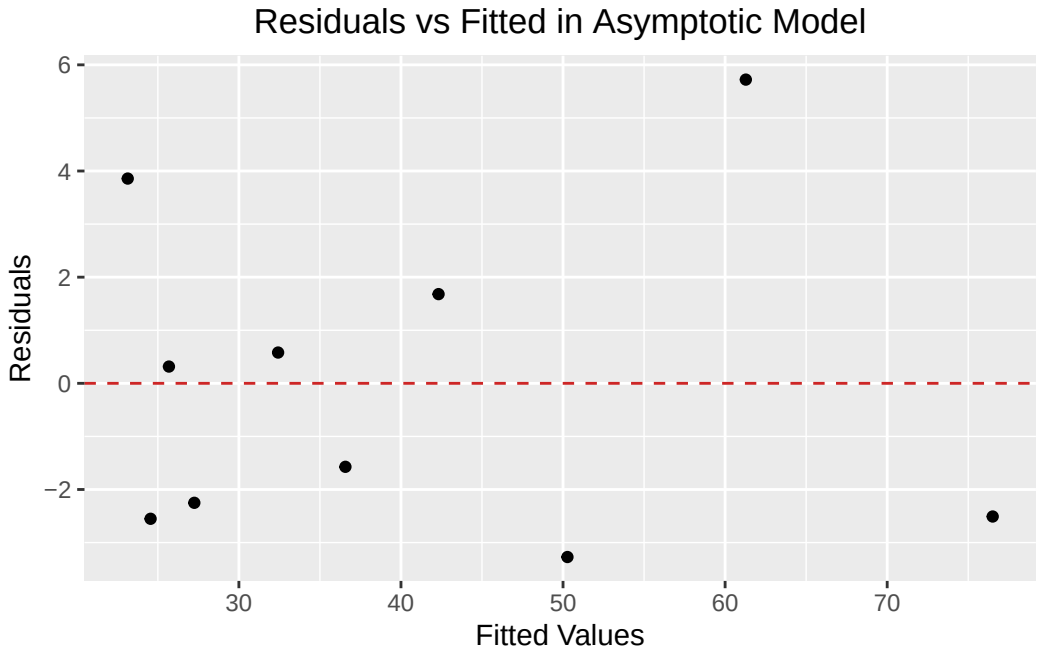
ggplot(res_vs_fit_exponential_snf, aes(x = Fitted, y = Residuals)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "forestgreen", linetype = "dashed") +
  labs(title = "Residuals vs Fitted in Exponential Decay Model (Session Number Used)", x = "
  theme(plot.title = element_text(hjust = 0.5))
```




```r
# Asymptotic

res_vs_fit_asymptotic_snf <- data.frame(
  Fitted = fitted(model_asymptotic_snf),
  Residuals = residuals(model_asymptotic_snf)
)

ggplot(res_vs_fit_asymptotic_snf, aes(x = Fitted, y = Residuals)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "firebrick3", linetype = "dashed") +
  labs(title = "Residuals vs Fitted in Asymptotic Model", x = "Fitted Values", y = "Residuals
  theme(plot.title = element_text(hjust = 0.5))
```




```
# ANOVA between nested models

anova(model_exponential_snf, model_asymptotic_snf)
```

```
Analysis of Variance Table

Model 1: Participants ~ a * exp(b * Session_Number)
Model 2: Participants ~ c + a * exp(b * Session_Number)
  Res.Df Res.Sum Sq Df Sum Sq F value   Pr(>F)
1      8    274.784
2      7     81.935  1 192.85  16.476 0.004815 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```